§ Definition of Continuity A function f is continuous at a point a if...

§ Important Theorem A function f is continuous at a point a if for every epsilon > 0, there exists a delta > 0 such that...

This also points to Important Theorem on page 1.

The Important Theorem provides the mathematical foundation.

Sieve of Eratosthenes (1–100)

Mathematical description. Fix $n = 100$. Let $S = \{2, 3, \ldots, n\}$. For each integer p defined as the smallest element of S not yet processed, declare p *prime* and remove from S every integer of the form kp with $k \geq 2$. Continue this procedure while $p^2 \leq n$. The elements of S that remain after this process are precisely the prime numbers $\leq n$.

Algorithm (short).

1. Initialize the list $2, 3, \ldots, n$.
2. For each current smallest unprocessed p (starting at $p = 2$), remove all multiples $2p, 3p, 4p, \ldots$ up to n.
3. Stop once $p^2 > n$ (i.e. $p > \lfloor \sqrt{n} \rfloor$). Remaining numbers are primes.

Note. 1 is neither prime nor composite.

	2	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

Primes The primes ≤ 100 are:

$$\mathbf{2},\ \mathbf{3},\ \mathbf{5},\ \mathbf{7},\ \mathbf{11},\ \mathbf{13},\ \mathbf{17},\ \mathbf{19},\ \mathbf{23},\ \mathbf{29}, \mathbf{31},\ \mathbf{37},\ \mathbf{41},\ \mathbf{43},\ \mathbf{47},\ \mathbf{53},\ \mathbf{59},\ \mathbf{61},\ \mathbf{67},\ \mathbf{71}, \mathbf{73},\ \mathbf{79},\ \mathbf{83},\ \mathbf{89},\ \mathbf{97}.$$

L1: Divisibility

Introduction

Elementary number theory is the study of arithmetic properties of the integers, especially divisibility properties of the integers $\mathbb{Z}$. Many classical results originate in Euclid's *Elements* (e.g., Euclid's proof of the infinitude of primes, the Euclidean algorithm, perfect numbers, Pythagorean triples).

Definition 0.1. Let $a, b \in \mathbb{Z}$. We say a *ldiv* b, and write $a \mid b$, if there exists an integer $c \in \mathbb{Z}$ such that

$$b = ac.$$

If no such integer c exists we write a
$NDIVb$.

Remark 0.1. When $a \mid b$ with $a \neq 0$, the integer $c = b/a$ is the exact (integer) quotient. The symbols a/b or b/a in plain text denote rational numbers (when the division is performed in $\mathbb{Q}$), which is a different notion from the divisibility relation $a \mid b$.

Example 0.1.

1. $3 \mid 6$, since $6 = 3 \cdot 2$.
2. $3 \mid -6$, since $-6 = 3 \cdot (-2)$.
3. For any $a \in \mathbb{Z}$ we have $a \mid 0$, because $0 = a \cdot 0$.
4. $0 \mid a$ only if $a = 0$. Indeed, $0 \mid a$ would require $a = 0 \cdot c = 0$ for some integer c.

Proposition 0.1 (Transitivity of divisibility)**.** *If $a, b, c \in \mathbb{Z}$ and $a \mid b$ and $b \mid c$, then $a \mid c$.*

Proof. Write $b = ae$ and $c = bf$ for some integers e, f. Then

$$c = bf = (ae)f = a(ef),$$

so $a \mid c$. $\square$

Proposition 0.2. *Let $a, b, c, m, n \in \mathbb{Z}$. If $c \mid a$ and $c \mid b$, then for all integers m, n we have*

$$c \mid (ma + nb).$$

Proof. Since $c \mid a$ and $c \mid b$, there exist $e, f \in \mathbb{Z}$ with $a = ce$ and $b = cf$. Thus

$$ma + nb = m(ce) + n(cf) = c(me + nf),$$

and therefore $c \mid (ma + nb)$. $\square$

The greatest integer (floor) function

Definition 0.2. For $x \in \mathbb{R}$, the *greatest integer function* (floor) is denoted $\lfloor x \rfloor$ and defined as the greatest integer less than or equal to x.

Example 0.2. If $a \in \mathbb{Z}$ then $\lfloor a \rfloor = a$. Also

$$\lfloor 5/3 \rfloor = 1, \qquad \lfloor \pi \rfloor = 3, \qquad \lfloor -5/3 \rfloor = -2, \qquad \lfloor -\pi \rfloor = -4.$$

Lemma 0.1. *For every $x \in \mathbb{R}$,*

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1.$$

Equivalently,

$$x - 1 < \lfloor x \rfloor \leq x.$$

Proof. By definition $\lfloor x \rfloor$ is an integer and it is the greatest integer not exceeding x, so $\lfloor x \rfloor \leq x$. If $x \geq \lfloor x \rfloor + 1$ then $\lfloor x \rfloor + 1$ would be an integer $\leq x$ larger than $\lfloor x \rfloor$, contradicting maximality of $\lfloor x \rfloor$. Therefore $x < \lfloor x \rfloor + 1$. $\square$

Division algorithm and remainders

When an integer (the divisor) is divided into another integer (the dividend), one obtains an integer quotient and an integer remainder.

Theorem 0.1 (Division Algorithm)**.** *Let $a, b \in \mathbb{Z}$ with $b > 0$. Then there exist unique integers q and r such that*

$$a = bq + r \qquad \textit{and} \qquad 0 \le r < b.$$

Here q is called the quotient *and r the* remainder*.*

Existence. Let $q := \lfloor a/b \rfloor$ and set $r := a - bq$. Clearly $a = bq + r$. By Lemma 0.1 applied to a/b we have

$$\lfloor a/b \rfloor \le \frac{a}{b} < \lfloor a/b \rfloor + 1,$$

so

$$0 \le a - b\lfloor a/b \rfloor < b,$$

i.e. $0 \le r < b$. □

Uniqueness. Suppose $a = bq_1 + r_1 = bq_2 + r_2$ with $0 \le r_1, r_2 < b$. Subtracting gives

$$0 = b(q_1 - q_2) + (r_1 - r_2),$$

so

$$b(q_1 - q_2) = r_2 - r_1.$$

Hence $bldiv(r_2 - r_1)$. But $-(b-1) \le r_2 - r_1 \le b - 1$, so the only multiple of b in that range is 0. Therefore $r_2 - r_1 = 0$, so $r_1 = r_2$, and consequently $q_1 = q_2$. This proves uniqueness. □

Example 0.3. Take $a = -5$ and $b = 3$. Then

$$q = \lfloor a/b \rfloor = \lfloor -5/3 \rfloor = -2, \qquad r = a - bq = -5 - 3(-2) = 1.$$

So $-5 = 3 \cdot (-2) + 1$ with $0 \le 1 < 3$.

Remark 0.2. Although one might informally write $-5 = 3 \cdot (-1) + (-2)$, this pair $(q, r) = (-1, -2)$ does satisfy $a = bq + r$ but *does not* satisfy the required inequality $0 \le r < b$. The Division Algorithm fixes a unique remainder in the interval $[0, b)$.

Parity

Definition 0.3. An integer $n \in \mathbb{Z}$ is *even* if there exists $k \in \mathbb{Z}$ such that $n = 2k$. An integer n is *odd* if there exists $k \in \mathbb{Z}$ such that $n = 2k + 1$.

Example 0.4. The set of even integers is

$$\{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\},$$

and the set of odd integers is

$$\{\dots, -5, -3, -1, 1, 3, 5, \dots\}.$$

L2: Prime Numbers

Definition 0.4. Let $p \in \mathbb{Z}$ with $p > 1$. We say p is *prime* if its only positive divisors are 1 and p. An integer $n > 1$ which is not prime is called *composite.*

Remark 0.3. The integer 1 is neither prime nor composite. (We will give a conceptual reason for this later in the course; it is tied to the uniqueness of prime factorization.)

Example 0.5. The primes between 2 and 50 are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.$$

Observe that every prime except 2 is odd.

Lemma 0.2. *Every integer $n > 1$ has at least one prime divisor.*

Proof. Suppose, toward contradiction, that there exists $n > 1$ with no prime divisor. By the well-ordering principle, take the smallest such n. Since n has no prime divisors, n cannot be prime, so n is composite: there exist integers a, b with $1 < a < n$, $1 < b < n$, and $n = ab$. By minimality of n, the factor a (being > 1 and $< n$) must have a prime divisor p. Then $pldiva$ and $aldivn$, so by transitivity $pldivn$, contradicting the assumption. Hence every $n > 1$ has a prime divisor. □

Theorem 0.2 (Euclid)**.** *There are infinitely many prime numbers.*

Proof. Assume, for contradiction, that there are only finitely many primes $p_1, p_2, \dots, p_k$. Consider

$$N := p_1 p_2 \cdots p_k + 1.$$

By the previous lemma N has a prime divisor p. This p cannot be any of $p_1, \dots, p_k$ because dividing N by any p_i gives remainder 1. Thus p is a prime not in the list, contradiction. Therefore primes are infinite in number. □

Proposition 0.3. *Let n be a composite integer. Then n has a prime divisor p with $p \le \sqrt{n}$.*

Proof. Write $n = ab$ with integers $1 < a \le b < n$. Then $a \le \sqrt{n}$ (otherwise $ab > \sqrt{n} \cdot \sqrt{n} = n$). By the lemma, a has a prime divisor p, so $pldiva$ and hence $pldivn$. Since $p \le a \le \sqrt{n}$, the conclusion follows. □

Remark 0.4. Proposition above underlies the *trial division* primality test and is the theoretical basis for the Sieve of Eratosthenes: to decide whether n is composite, it suffices to check divisibility by primes $\le \sqrt{n}$.

Example 0.6 (Sieve idea)**.** To find all primes ≤ 50, it suffices to remove multiples of primes $\le \sqrt{50} \approx 7.07$, i.e., delete multiples of $2, 3, 5, 7$ (except the primes themselves). Remaining integers are the primes listed earlier.

Proposition 0.4. *For any positive integer n, there exist n consecutive composite positive integers.*

Proof. Consider the integers

$$(n+1)! + 2,\ (n+1)! + 3,\ \dots,\ (n+1)! + (n+1).$$

For any $2 \le i \le n+1$, we have $ildiv(n+1)!$ and $ildivi$, hence $ildiv\big((n+1)! + i\big)$. Thus each term is composite (it has the factor i). □

Remark 0.5. Pairs $(p, p+2)$ of primes are called *twin primes* (e.g. $(3, 5), (5, 7), (11, 13), \dots$). The Twin Prime Conjecture asserts there are infinitely many such pairs; it remains open, though there has been substantial progress on bounded prime gaps (Zhang and subsequent work).

Prime number distribution and famous conjectures

Definition 0.5. For $x > 0$, let

$$\pi(x) := \#\{p \text{ prime} : p \leq x\},$$

the prime-counting function.

Theorem 0.3 (Prime Number Theorem)**.**

$$\lim_{x\to\infty} \frac{\pi(x)}{x/\log x} = 1.$$

Remark 0.6. The Prime Number Theorem (PNT) states that $\pi(x) \sim x/\log x$. Its proof uses complex analysis and was independently obtained by Hadamard and de la Vallée Poussin in 1896. A proof is beyond the scope of this course, but the statement explains the average density of primes among large integers.

Special families of primes

Definition 0.6. A *Mersenne* prime is a prime of the form $2^p - 1$ where p is prime.

Example 0.7. The first few Mersenne primes are

$$3 = 2^2 - 1, \quad 7 = 2^3 - 1, \quad 31 = 2^5 - 1, \quad 127 = 2^7 - 1, \quad 8191 = 2^{13} - 1.$$

Note $2^{11} - 1 = 2047 = 23 \cdot 89$ is composite (so not all numbers $2^p - 1$ are prime).

Definition 0.7. A *Fermat* prime is a prime of the form $2^{2^n} + 1$ for some integer $n \geq 0$.

Example 0.8. The Fermat numbers for $n = 0, 1, 2, 3, 4$ give

$$3, 5, 17, 257, 65537,$$

all prime. However the next, $F_5 = 2^{2^5} + 1 = 2^{32} + 1 = 4294967297$, is composite; Euler showed that $641ldivF_5$, so not all Fermat numbers are prime.

Remark 0.7. Fermat conjectured that all numbers of the form $2^{2^n} + 1$ are prime; Euler disproved this by showing F_5 is composite. It is unknown whether there are infinitely many Fermat primes; current evidence suggests only finitely many are prime (only the five above are known to be prime).

Conjecture 0.1 (Goldbach, 1742)**.** *Every even integer greater than* 2 *is the sum of two (not necessarily distinct) primes.*

Remark 0.8. Goldbach's conjecture has been verified by computation up to very large bounds, but a proof remains open. Hardy and Littlewood proposed an asymptotic formula for representations of even numbers as sums of two primes, but the full conjecture is unresolved.

Conjecture 0.2 (Hardy–Littlewood quadratic prime conjecture)**.** *There are infinitely many primes of the form* $n^2 + 1$ *for integer* $n \geq 0$.

Remark 0.9. Note $n^2 - 1 = (n+1)(n-1)$ is composite for $n > 2$, so this form yields no new large primes for $n > 2$. Factorizations are a central tool for ruling out primality in many forms.

L3: Greatest Common Divisors

Definition 0.8. Let $a, b \in \mathbb{Z}$, not both zero. The *set of common divisors* of a and b is

$$\Delta := \{c \in \mathbb{Z} : cldiva \text{ and } cldivb\}.$$

Remark 0.10. If a and b are not both zero then Δ is nonempty (since $\pm 1 \in \Delta$) and finite (every common divisor ldiv $|a|$ or $|b|$, and any nonzero integer has only finitely many divisors). Hence the greatest element of Δ exists and is positive.

Definition 0.9. Let $a, b \in \mathbb{Z}$, not both zero. The *greatest common divisor* of a and b, denoted $\gcd(a, b)$, is the greatest positive integer d such that $dldiva$ and $dldivb$. If $\gcd(a, b) = 1$ we say a and b are *relatively prime.*

Remark 0.11. The quantity $\gcd(0, 0)$ is undefined. Also, if $\gcd(a, b) = d$ then

$$\gcd(-a, b) = \gcd(a, -b) = \gcd(-a, -b) = d.$$

Thus we may restrict many computations to nonnegative integers without loss of generality.

Example 0.9. Divisors of 24: $\pm1, \pm2, \pm3, \pm4, \pm6, \pm8, \pm12, \pm24$.
Divisors of 60: $\pm1, \pm2, \pm3, \pm4, \pm5, \pm6, \pm10, \pm12, \pm15, \pm20, \pm30, \pm60$.
Hence $\gcd(24, 60) = 12$.

Remark 0.12. If $a \in \mathbb{Z}$ with $a \neq 0$ then $\gcd(a, 0) = |a|$.

Proposition 0.5. *Let $a, b \in \mathbb{Z}$ and let $\gcd(a, b) = d$. Then $\gcd(a/d,\ b/d) = 1$.*

Proof. Let $d' = \gcd(a/d, b/d)$. Then $d'ldiva/d$ and $d'ldivb/d$, so there exist $e, f \in \mathbb{Z}$ with $a/d = d'e$ and $b/d = d'f$. Hence $a = d'de$ and $b = d'df$, so $d'd$ ldiv both a and b. By maximality of d as the gcd we must have $d'd \leq d$, so $d' = 1$. $\square$

Proposition 0.6 (Bézout's characterization of the gcd)**.** *Let $a, b \in \mathbb{Z}$, not both zero. Then*

$$\gcd(a, b) = \min\{ma + nb : m, n \in \mathbb{Z},\ ma + nb > 0\}.$$

Moreover, there exist integers m_0, n_0 such that

$$\gcd(a, b) = m_0 a + n_0 b.$$

Proof. Let $S = \{ma + nb : m, n \in \mathbb{Z},\ ma + nb > 0\}$. Since a and b are not both zero, $S \neq \varnothing$ (take $m = 1, n = 0$ or $m = -1, n = 0$, etc.), and by the well-ordering principle S has a least positive element; denote it d. So $d = m'a + n'b$ for some $m', n' \in \mathbb{Z}$.

We show $dldiva$ and $dldivb$. Apply the division algorithm to write $a = dq + r$ with $0 \leq r < d$. Then

$$r = a - dq = a - q(m'a + n'b) = (1 - qm')a + (-qn')b,$$

so $r \in S \cup \{0\}$. Since $0 \leq r < d$ and d is the least positive element of S, we must have $r = 0$. Thus $dldiva$. Similarly $dldivb$.

Finally, if c is any common divisor of a and b, then $cldiv(ma + nb)$ for any integers m, n, hence $cldivd$. Therefore $c \leq d$. Thus d is the greatest common divisor. This proves the claim. $\square$

Remark 0.13. The fact that the gcd is an integral linear combination of a and b is commonly referred to as Bézout's identity. In the special case $\gcd(a, b) = 1$ the identity shows there exist integers m, n with $ma + nb = 1$; any common divisor of a and b must then divide 1, so it is 1.

Definition 0.10. For integers $a_1, \ldots, a_k$ not all zero, their greatest common divisor $\gcd(a_1, \ldots, a_k)$ is the largest positive integer dividing each a_i. If $\gcd(a_1, \ldots, a_k) = 1$ we say the a_i are *relatively prime.* If every pair $\gcd(a_i, a_j) = 1$ for $i \neq j$ we say they are *pairwise relatively prime.*

Example 0.10. The common divisors of $24, 60, 30$ are $\pm1, \pm2, \pm3, \pm6$, so $\gcd(24, 60, 30) = 6$. Note that pairwise relatively prime integers are relatively prime, but the converse need not hold.

The Euclidean Algorithm

Lemma 0.3. *Let $a, b \in \mathbb{Z}$ with $a \geq b > 0$. If $a = bq + r$ with integers q, r and $0 \leq r < b$, then*

$$\gcd(a, b) = \gcd(b, r).$$

Proof. If c ldiv a and b, then $cldiv(a - qb) = r$, so c ldiv b and r. Conversely, if c ldiv b and r then $cldiv(qb+r) = a$. Thus the set of common divisors of $\gcd(a, b)$ equals the set of common divisors of $\gcd(b, r)$, so their greatest elements are equal. □

Theorem 0.4 (Euclidean algorithm)**.** *Let $a, b \in \mathbb{Z}$ with $a \geq b > 0$. Perform repeated division:*

$$\begin{aligned} a &= bq_1 + r_1, \quad 0 \leq r_1 < b, \\ b &= r_1 q_2 + r_2, \quad 0 \leq r_2 < r_1, \\ r_1 &= r_2 q_3 + r_3, \quad 0 \leq r_3 < r_2, \\ &\vdots \end{aligned}$$

This process terminates: for some $n \geq 1$, $r_n = 0$. Then

$$\gcd(a, b) = \begin{cases} b & \text{if } n = 1, \\ r_{n-1} & \text{if } n > 1. \end{cases}$$

Proof. The remainders form a strictly decreasing sequence of nonnegative integers:

$$b > r_1 > r_2 > \cdots \geq 0.$$

Such a sequence cannot be infinite, so for some n we must have $r_n = 0$. Repeated application of Lemma 0.3 yields

$$\gcd(a, b) = \gcd(b, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_{n-1}, 0) = r_{n-1}.$$

If $n = 1$ then $r_1 = 0$ already and $\gcd(a, b) = b$. □

Example 0.11. Compute $\gcd(803, 154)$ via the Euclidean algorithm:

$$\begin{aligned} 803 &= 154 \cdot 5 + 33, \\ 154 &= 33 \cdot 4 + 22, \\ 33 &= 22 \cdot 1 + 11, \\ 22 &= 11 \cdot 2 + 0. \end{aligned}$$

Hence $\gcd(803, 154) = 11$.

Back-substitution to obtain Bézout coefficients

We can express the gcd as a linear combination of the original numbers by back-substitution.

From the third nonzero remainder:

$$11 = 33 - 22 \cdot 1.$$

Substitute $22 = 154 - 33 \cdot 4$ (from the second line):

$$11 = 33 - (154 - 33 \cdot 4) = 33 \cdot 5 - 154.$$

Substitute $33 = 803 - 154 \cdot 5$ (from the first line):

$$11 = (803 - 154 \cdot 5) \cdot 5 - 154 = 803 \cdot 5 - 154 \cdot 26.$$

Thus

$$11 = 5 \cdot 803 + (-26) \cdot 154,$$

so $\gcd(803, 154) = 11$ is a linear combination of 803 and 154. (These coefficients are not unique.)

L4: The Fundamental Theorem of Arithmetic

Lemma 0.4 (Euclid's lemma)**.** *Let $a, b, p \in \mathbb{Z}$ with p prime. If $pldivab$, then $pldiva$ or $pldivb$.*

Proof. Suppose p
$NDIVa$. Then $\gcd(a, p) = 1$. By Bézout's identity there exist integers m, n such that

$$ma + np = 1.$$

Multiply both sides by b to obtain

$$mab + npb = b.$$

Since $pldivab$, the term mab is divisible by p, and clearly npb is divisible by p, hence the left-hand side is divisible by p. Therefore $pldivb$. This proves the lemma. □

Corollary 0.1. *Let $a_1, \ldots, a_n, p \in \mathbb{Z}$ with p prime. If $pldiv(a_1 a_2 \cdots a_n)$ then $pldiva_i$ for some i.*

Proof. Prove by induction on n. The base case $n = 1$ is trivial. The case $n = 2$ is Lemma 0.4. For the inductive step, assume the statement for $n = k$ and suppose $pldiv(a_1 \cdots a_k a_{k+1})$. By Lemma 0.4, either $pldiva_{k+1}$ (done) or $pldiv(a_1 \cdots a_k)$, in which case the induction hypothesis yields $pldiva_i$ for some $i \leq k$. □

Theorem 0.5 (Fundamental Theorem of Arithmetic)**.** *Every integer $n > 1$ can be written as*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r},$$

where the p_i are distinct primes and the α_i are positive integers. Moreover this factorization is unique up to the ordering of the prime factors.

Existence. Assume, for contradiction, that there exists $n > 1$ having no prime factorization. By the well-ordering principle let N be the least such integer. Then N is not prime (otherwise N itself is a one-term factorization), so $N = ab$ with $1 < a, b < N$. By minimality of N, both a and b have prime factorizations; hence so does $N = ab$, contradiction. Therefore every $n > 1$ has a prime factorization. □

Uniqueness. Suppose

$$p_1^{\alpha_1} \cdots p_r^{\alpha_r} = q_1^{\beta_1} \cdots q_s^{\beta_s},$$

where the p_i's and the q_j's are primes and each list is in increasing order. Take p_1 (the smallest p). Since p_1 ldiv the right-hand side, Corollary 0.1 shows p_1 equals one of the q_j; by ordering we may assume $p_1 = q_1$. Cancel the common power $p_1^{\min(\alpha_1, \beta_1)}$ from both sides to reduce to a similar equality with strictly smaller total exponent sum. Repeating this cancellation shows the multisets of primes on both sides coincide; hence $r = s$, $p_i = q_i$ for all i, and comparing exponents yields $\alpha_i = \beta_i$ for each i. This proves uniqueness. □

Example 0.12. Prime factorization: $756 = 2^2\, 3^3\, 7^1$.

Definition 0.11. For $a, b \in \mathbb{Z}$, $a, b > 0$, the *least common multiple* $\operatorname{lcm}(a, b)$ is the least positive integer m such that $aldivm$ and $bldivm$.

Remark 0.14. Equivalently, the set $T = \{c \in \mathbb{Z} : c > 0,\ aldivc,\ bldivc\}$ is nonempty (contains ab) and by the well-ordering principle has a least element $\operatorname{lcm}(a, b)$.

Proposition 0.7. *Let $a, b > 1$ and write their prime factorizations using the same distinct primes*

$$a = \prod_{i=1}^{n} p_i^{\alpha_i}, \qquad b = \prod_{i=1}^{n} p_i^{\beta_i},$$

where $\alpha_i, \beta_i \geq 0$ (possibly zero). Then

$$\gcd(a, b) = \prod_{i=1}^{n} p_i^{\min\{\alpha_i, \beta_i\}}, \qquad \operatorname{lcm}(a, b) = \prod_{i=1}^{n} p_i^{\max\{\alpha_i, \beta_i\}}.$$

Proof. Immediate from the definitions of gcd and lcm and the uniqueness of prime exponents in the prime factorizations. □

Example 0.13. Factorizations:

$$756 = 2^2 3^3 5^0 7^1, \qquad 2205 = 2^0 3^2 5^1 7^2.$$

Thus

$$\gcd(756, 2205) = 2^0 3^2 5^0 7^1 = 9 \cdot 7 = 63,$$

and

$$\operatorname{lcm}(756, 2205) = 2^2 3^3 5^1 7^2 = 26460.$$

Lemma 0.5. *For real numbers x, y,*

$$\min\{x, y\} + \max\{x, y\} = x + y.$$

Theorem 0.6. *For positive integers a, b,*

$$\gcd(a, b)\ \operatorname{lcm}(a, b) = ab.$$

Proof. Write $a = \prod p_i^{\alpha_i}$ and $b = \prod p_i^{\beta_i}$ as in Proposition 0.7. Then by that proposition and Lemma 0.5,

$$\gcd(a, b)\ \operatorname{lcm}(a, b) = \prod_i p_i^{\min(\alpha_i, \beta_i) + \max(\alpha_i, \beta_i)} = \prod_i p_i^{\alpha_i + \beta_i} = \Big(\prod_i p_i^{\alpha_i}\Big)\Big(\prod_i p_i^{\beta_i}\Big) = ab.$$

This proves the identity. (The cases involving 1 are immediate.) □

Corollary 0.2. *For positive integers a, b,* $\operatorname{lcm}(a, b) = ab$ *if and only if* $\gcd(a, b) = 1$.

Proof. Using Theorem 0.6, $ab = \gcd(a, b)\ \operatorname{lcm}(a, b)$. Hence $\operatorname{lcm}(a, b) = ab$ iff $\gcd(a, b) = 1$. □

Remark 0.15. In practice, for large integers it is usually more efficient to compute $\gcd(a, b)$ via the Euclidean algorithm and then compute

$$\operatorname{lcm}(a, b) = \frac{ab}{\gcd(a, b)}$$

rather than factoring large numbers.